

Assembly Evaluation Reflection

1. Relevance of Metrics

The goal of this project is to evaluate the quality of multiple *Salmonella* serovars bacterial genome assemblies using assembly evaluation pipeline(s). Quality of genome assembly is critical, since analyses, such as genomics comparing, phylogenetics, and studies on gene presence/absence heavily depend on accuracy and assembly completeness. Poor assembly quality can introduce to errors (missing data, gene absences, and/or biased conclusions).

To evaluate the quality of assemblies, analyzing the metrics measuring system representing the three core principles of assembly assessment: contiguity, completeness, and correctness is required. These metrics include total assembly length, number of contigs, N50 values, and describes how continuous or fragmented each genome assembly are. BUSCO (Benchmarking Universal Single-Copy Orthologs) uses the bacteria_odb10 dataset to assess the expected single-copy genes in the bacterial genomes and calculate completeness.

BUSCO measures completeness by classifying genes as Complete (C), Single-copy (S), Duplicated (D), Fragmented (F) and/or Missing (M). A high-quality bacterial assembly typically contains more than 95% complete BUSCO genes, indicating that nearly all the expected orthologs are present. Complete and single-copy genes represent ideal assembly performance. Duplicated genes may indicate assembly redundancy, while fragmented genes suggest assembly breaks across contigs. Missing genes generally indicate incomplete sequencing or assembly gaps.

While BUSCO primarily measures completeness, high BUSCO scores indirectly support assembly correctness since genes must be intact to be classified as complete. However, BUSCO does not guarantee accuracy. Assemblies may still contain inversions, translocations, or other misassemblies despite high completeness scores. Therefore, BUSCO results must be interpreted alongside contiguity metrics such as N50 and contig count.

Requirements to classify assemblies as “good”:

- High N50 values relative to genome size
- Low numbers of contigs (minimal fragmentation)
- BUSCO completeness $\geq 95\%$
- Low fragmented (F) and missing (M) percentages
- Minimal duplication (D), indicating limited redundancy

Together, these metrics provide a comprehensive evaluation of the three C's: contiguity, completeness, and inferred correctness.

2. Assembly Evaluation Results

Summary of the 3 C's

A summary (included in the GitHub repository as `assembly_evaluation/assembly_stats.tsv`) illustrates N50 values and BUSCO completeness percentages across all thirteen genomes. Overall, most assemblies demonstrate a strong contiguity and high completeness, with a small subset showing clear fragmentation and reduced completeness.

BUSCO completeness ranged from approximately 67.7% to 100%. N50 values ranged from approximately 103 kb in the most fragmented assembly to over 4.7 Mb in nearly complete single contig assemblies. Most genomes cluster above 98% completeness with relatively few contigs, indicating strong overall assembly performance.

Contiguity

Most assemblies exhibited strong contiguity. Several genomes consisted of only 1–5 contigs with N50 values approaching total genome length, indicating nearly complete chromosome assemblies. These assemblies are structurally robust and likely suitable for downstream structural comparisons.

However, a subset of serovars showed substantially higher contig counts (up to 113 contigs) and much lower N50 values (~103 kb to ~370 kb). These genomes are more fragmented, suggesting unresolved repetitive regions or lower sequencing quality. While still usable for certain analyses, these assemblies may limit the reliability of structural comparisons such as synteny analysis.

Completeness

BUSCO analysis revealed high completeness across most assemblies. Many genomes exhibited $\geq 98\%$ complete BUSCO genes, and two reached 100% completeness with no fragmented or missing orthologs. These results indicate strong completeness and high recovery of conserved gene content.

However, a small subset of genomes showed lower completeness (approximately 67–81%) with elevated missing gene percentages (~17–26%). These assemblies are less complete and lack portions of expected conserved genes. Such deficiencies could influence gene presence/absence analyses and inflate estimates of gene loss.

Duplication rates were generally low across assemblies, suggesting minimal redundancy and few over-assembly artifacts.

Correctness

Correctness is assessed indirectly through combined interpretation of contiguity and completeness metrics. Assemblies with high N50 values, low contig counts, and high BUSCO completeness are unlikely to contain major structural errors. The presence of intact single-copy conserved genes supports structural integrity at the gene level.

However, no direct misassembly detection tools or read-mapping validation steps were performed in this analysis. Therefore, while gene-level correctness appears strong for most genomes, structural accuracy cannot be fully confirmed. Assemblies with high fragmentation and lower completeness may be more prone to misassemblies or unresolved repeat regions.

To strengthen correctness assessment in future analyses, incorporating read alignment validation or misassembly detection tools would provide additional confirmation of structural integrity.

Assemblies of Concern and Downstream Suitability

Most assemblies are suitable for downstream comparative genomic analyses. Genomes with $\geq 98\%$ BUSCO completeness and strong N50 values represent high-quality assemblies appropriate for gene presence/absence studies and phylogenetic analysis.

However, assemblies with $\sim 67\text{--}81\%$ BUSCO completeness and high contig counts represent lower-quality genomes. These assemblies may introduce bias into downstream analyses by artificially increasing missing gene counts or reducing structural reliability. If the analysis requires high-confidence gene content comparisons, it is best to consider excluding the most fragmented assemblies (particularly those below $\sim 90\%$ BUSCO completeness) or clearly acknowledging their limitations.

Overall, many assemblies meet quality expectations for bacterial genome analysis, with only a small subset requiring caution.

3. Tool Execution

The selected tools performed as expected. Assembly summary statistics provided clear structural metrics, and BUSCO offered biologically meaningful completeness assessment rather than purely structural measurements. BUSCO outputs were straightforward to interpret and allowed efficient comparison across genomes.

During the workflow, challenges arose related to file organization and GitHub tracking of large BUSCO output directories. These were logistical rather than analytical issues and were resolved by refining file management strategies. The tools themselves executed reliably within the conda environment.

There were no unexpected surprises in the results. The high completeness of most genomes aligned with expectations for bacterial assemblies. However, the variability in contiguity among serovars highlighted the importance of evaluating assembly quality before conducting downstream analyses.

For future reproduction, repeat this process on a new dataset with the use of the same core tools and settings. BUSCO remains a standard of metric for completeness assessment, and N50 and contig counts remain essential structural metrics. However, for future use adding a direct correctness validation step, such as read mapping or a misassembly detection tool, to strengthen confidence in structural accuracy.

Conclusion

The assembly evaluation pipeline demonstrated that most genomes in this dataset are high quality, exhibiting strong contiguity and high biological completeness. A small subset of assemblies showed increased fragmentation and reduced completeness, warranting cautious interpretation or potential exclusion from certain downstream analyses.

This project reinforced the importance of evaluating genome assemblies using multiple complementary metrics. Considering contiguity, completeness, and correctness together ensures that downstream comparative analyses are based on reliable and biologically meaningful data. Careful assembly evaluation is essential for maintaining the integrity of bioinformatic conclusions.